

RWANDA TVET BOARD
COMPETENCY-BASED ASSESSMENT
[SUMMATIVE ASSESSMENT]

Program: SOFTWARE DEVELOPMENT | **Level:** 4 | **Total Marks:** 100

Module: Backend Application Development Using Node JS | **Date:** 2026-05-26 | **Time:** 3 Hours

Name: _____ Index No: _____ Class: _____

SECTION B: MULTIPLE CHOICE

Choose ONE correct answer and circle the letter.

1 How is Unit Testing used when managing Backend Application? **(2 marks)**

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- A. Unit Testing
- B. Authorization
- C. Nodemon
- D. Node.js

Answer: _____

2 What is the significance of Node.js when developing RESTFUL APIs with Node JS? **(2 marks)**

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- A. Chai
- B. Request body
- C. Node.js
- D. Continuous deployment

Answer: _____

3 What is the significance of RESTful API when securing Backend Application? **(2 marks)**

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- A. Error handling middleware
- B. MySQL
- C. Authentication middleware
- D. RESTful API

Answer: _____

4 How is Supertest used when testing Backend Application? **(2 marks)**

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- A. Node.js application deployment
- B. MySQL
- C. HTTP (Hypertext Transfer Protocol)
- D. Supertest

Answer: _____

SECTION C: MATCHING

Match each term in Column A with Column B.

5 Match each term in Column A with its correct description in Column B. **(4 marks)**

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Column A (Terms)	Column B (Descriptions)
1. Node.js	A. is an open-source JavaScript runtime environment that allows
2. A backend application	B. is the server-side component of a web application
3. An object in JavaScript	C. is a composite data type that stores key-value pairs.
4. Express.js	D. is a popular web application framework for Node.js. It simplifies the

Answers: 1. ____ 2. ____ 3. ____ 4. ____

6 Match each term in Column A with its correct description in Column B. (4 marks)

Column A (Terms)	Column B (Descriptions)
1. Hashing	A. is the process of converting input data (or a message) into a fixed-size
2. Node.js Package Manager (NPM)	B. is a powerful tool used for managing and
3. Express	C. is a web application framework for Node.js, designed to simplify the
4. Lodash	D. is a utility library that provides useful functions for common

Answers: 1. ____ 2. ____ 3. ____ 4. ____

7 Match each term in Column A with its correct description in Column B. (4 marks)

Column A (Terms)	Column B (Descriptions)
1. Mocha	A. is a popular JavaScript testing framework that runs on Node.js and in the
2. Chai	B. is a popular assertion library that works well with Mocha to write unit tests
3. TelaTech Ltd	C. is a company that develop websites for different institutions they have
4. Usability testing	D. is a technique used to evaluate a product or service by testing it

Answers: 1. ____ 2. ____ 3. ____ 4. ____

8 Match each term in Column A with its correct description in Column B. (4 marks)

Column A (Terms)	Column B (Descriptions)
1. Node.js application deployment	A. refers to the process of preparing, transferring,
2. Continuous deployment	B. is an automated process where changes in the codebase
3. IIS (Internet Information Services)	C. is a web server from Microsoft, and it can be
4. MySQL	D. is a popular relational database management system, and it will be used

Answers: 1. ____ 2. ____ 3. ____ 4. ____

SECTION D: OPEN-ENDED

Answer ALL questions. Write in the spaces provided.

- 9** Demonstrate how to use MongoDB when developing RESTFUL APIs with Node JS. **(10 marks)**
• Provide a step-by-step explanation.

- 10** Demonstrate how to use RESTful API when developing RESTFUL APIs with Node JS. **(6 marks)**
• Provide a step-by-step explanation.

- 11** Demonstrate how to use Middleware when developing RESTFUL APIs with Node JS. **(6 marks)**
1 Provide a step-by-step explanation.
• _____

- 12** Demonstrate how to use Node.js when securing Backend Application. Provide a **(6 marks)**
2 step-by-step explanation.
• _____

- 13** Demonstrate how to use JWT authentication when securing Backend Application. **(6 marks)**
3 Provide a step-by-step explanation.
• _____

- 14** Design a solution that uses Middleware to securing Backend Application. Explain your **(6 marks)**
4 design decisions.
• _____

- 15** Design a solution that uses Node.js to testing Backend Application. Explain your design **(6 marks)**
5 decisions.
• _____

- 16** Analyze the importance of REST API in testing Backend Application. Identify its **(6 marks)**
6 advantages and limitations.
• _____

- 17** Demonstrate how to use Unit Testing when testing Backend Application. Provide a **(6 marks)**
7 step-by-step explanation.
• _____

- 18** Design a solution that uses Node.js to managing Backend Application. Explain your **(6 marks)**
8 design decisions.
• _____

- 1** Analyze the importance of REST API in managing Backend Application. Identify its **(6 marks)**
9 advantages and limitations.

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- 2** Demonstrate how to use Integration Testing when managing Backend Application. **(6 marks)**
0 Provide a step-by-step explanation.

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END OF EXAMINATION | Summative | Backend Application Development Using Node JS | Total: 100 marks